

ID	Design driver	Engineering requirement	Mapped subsystem(s)
REQ-MOTION	Slow sagittal mobilization	Controlled dorsiflexion–plantarflexion under angular feedback.	ATM ACT CTRL
REQ-USE	Early rehabilitation before gait	Operate without walking or body-weight support.	TU FU EXP
REQ-FIT	Anthropometric variability	Provide adjustment at tibial and foot interfaces.	TU FU ATM
REQ-ALIGN	Ankle is not a fixed hinge	Provide practical geometric positioning of the measurement mechanism.	TU ATM
REQ-COMP	Secondary motion / residual misalignment	Tolerate limited out-of-plane motion without active control.	ATM
REQ-ACT	Bidirectional motion needs opposing tensile actions	Implement dorsiflexion- and plantarflexion-oriented SMA groups.	ACT
REQ-TRANS	Remote actuation with limited distal bulk	Provide flexible tensile transmission to tibial and foot structures.	TU FU ACT
REQ-SENSE	Closed-loop control needs a repeatable angle	Integrate non-contact relative-angle feedback.	ATM ELEC
REQ-CTRL	Opposing actuators require coordination	Provide tracking, directional allocation, safety supervision, and final gating.	ELEC CTRL
REQ-HEAT	SMA heating is faster than passive cooling	Reduce unnecessary heating and provide zero-command recovery opportunities.	ACT CTRL
REQ-EVAL	Engineering characterization precedes human evaluation	Support repeatable preliminary evaluation of the integrated system.	EXP

Subsystem colours are consistent with the system architecture: TU tibial unit; FU foot unit; ATM angle tracking; ACT SMA actuation/transmission; ELEC power/sensing electronics; CTRL embedded control/supervision; EXP experimental infrastructure.